Adaptive Morphological Post-processing for Enhanced Brain Segmentation Accuracy

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***Abstract*—Neurodegenerative diseases, like Alzheimer's disease, lead to progressive brain atrophy and cognitive decline. Magnetic resonance imaging is pivotal in diagnosing and tracking these conditions by enabling the segmentation of brain structures. Although manual segmentation is the gold standard, it is labor-intensive and variable, driving the push for automated solutions. This work introduces an adaptive morphological post-processing workflow, combining data-driven opening and 2D hole-filling techniques, to enhance brain segmentation. By addressing artifacts like protrusions, insulations, and small holes, the method improves segmentation quality, as demonstrated through quantitative (Dice coefficient, Hausdorff distance) and qualitative evaluations. Despite challenges, such as balancing artifact removal with preserving clinically relevant details, the workflow presents a robust, sustainable approach. Future work could focus on interdisciplinary collaboration to facilitate clinical integration, alongside rigorous validation on diverse datasets to ensure broad applicability and maximize impact.**

I. Introduction

Neurodegenerative diseases such as Alzheimer (AD), which affects up to 15% of individuals over the age of 65 and nearly half of those by age 85 [1], among others, significantly impact patients' quality of life. Early diagnosis and precise monitoring of these conditions are critical for improving treatment outcomes. Magnetic resonance imaging (MRI) plays a vital role in the diagnosis and analysis of brain diseases due to its ability to non-invasively provide high-resolution, medical images of brain structures.

In the field of Medical Image Analysis (MIA), the manual segmentation of magnetic resonance (MR) images has become a valuable tool for quantifying brain tissue changes associated with neurodegenerative diseases. This technique isolates key brain structures, including the Hippocampus and Grey Matter, enabling early diagnosis, treatment planning, and monitoring of disease progression.

Despite its advantages, manual segmentation remains labor-intensive, prone to inter-observer variability, and costly [2]. To address these limitations, machine learning-based automation has gained traction. By leveraging machine learning algorithms, such as Random Forest (RF) Classifiers, brain MR images can be segmented into meaningful categories — such as White Matter, Grey Matter, Hippocampus, Amygdala, and Thalamus — offering a faster, more consistent, and scalable solution for analyzing neurodegenerative diseases. However, such methods often present artifacted results which would not come up in manual segmentation such as protrusions, insulation and holes in the segmentation.

To address this issue, this paper presents a novel adaptive morphological post-processing step designed to improve segmentation quality. Post processing was tested on the output of a previously developed machine learning-based pipeline. The pipeline comprised key stages, including pre-processing (e.g. skull stripping, intensity normalization), image registration, feature extraction and finally segmentation by an RF classifier.

The hypothesis is that adaptive morphological post-processing will enhance the accuracy, in terms of established metrics, of automated brain region segmentation from MR images. Post-processing focuses on artifact removal (e.g. protrusions, insulations and holes) to improve segmentation quality and clarity, thereby enhancing the overall reliability of the outputs. Accurate segmentation is particularly critical for neuroimaging analyses, including studying brain structure, diagnosing diseases, treatment planning, and monitoring disease progression.

II. Methodologies

A. MRI Dataset

This study used 3 Tesla head MR images from the Human Connectome Project (HCP), comprising 30 unrelated healthy subjects [3]. For each subject, the dataset included T1-weighted (T1w) and T2-weighted (T2w) MR image volumes in native T1w space. Both modalities were defaced for anonymization, included bias field corrections, and were not skull-stripped. Ground truth label maps and brain masks were provided in the same space, alongside affine transformations for alignment to an atlas.

The ground truth labels were generated using FreeSurfer 5.3 and represent a "silver-standard" annotation. The segmentation consists of five labels: Grey Matter, White Matter, Amygdala, Hippocampus, and Thalamus. Additionally, the dataset included MR images and labels registered to the MNI152 atlas using nonlinear FNIRT. This provided standardized T1w and T2w atlas images, as well as an associated brain mask, facilitating spatial normalization and comparative analysis.

B. MIA Pipeline

The MIA pipeline serves as the foundational framework for the post-processing experiments. It was implemented in Python (v3.10.7) and consists of four main steps: (i) loading and pre-processing the training data, which includes normalization, skull stripping, and registration of T1w and T2w images along with their corresponding ground truth (GT) labels, as well as loading the atlas; (ii) training a RF classifier on the extracted feature matrices. The RF implementation uses the ‘scikit-learn‘ library (v1.4.2) with 20 estimators and a maximum depth of 50; (iii) predicting brain tissue labels for pre-processed test images using the trained RF classifier; and (iv) splitting the predicted output into label-specific binary images, applying post-processing operations to each label, and merging the processed labels to form the final segmentation output. The overall workflow of the pipeline is illustrated in Figure 1.

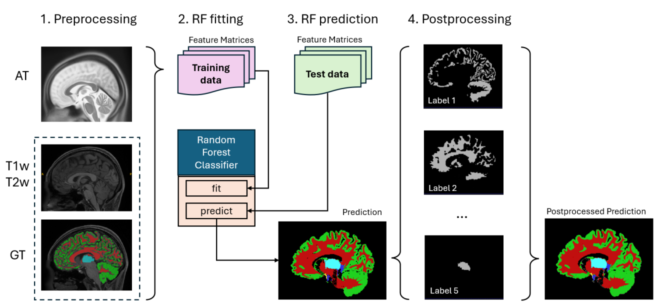

Fig. 1: Basic workflow of the MIA pipeline. AT = Atlas.

C. Post-processing Pipeline

The MIA pipeline up to the post-processing step (step iv) was established as preparatory work for this project, allowing the structure, functions, and training process to remain fixed. Specifically, the pre-processing and prediction steps were executed once, and the results were compressed and stored on disk using ‘joblib‘ (v1.4.0). The post-processing phase of the pipeline was encapsulated and slightly modified to load and decompress the stored data before applying the post-processing operations and evaluating the results. This restructured approach effectively created a standalone post-processing pipeline, enabling focused experimentation on improving the final segmentation outputs.

D. Morphological Operations

A preparatory review of established post-processing techniques for AI-driven MRI segmentation was conducted. Given the time constraints of the project and computational limitations of the available hardware, the selection process applied strict exclusion criteria, including implementability, computational efficiency, and compatibility with the existing pipeline. From this review, morphological filtering emerged as the foundation for the post-processing implementation. This technique, widely used in image processing, applies structural transformations to binary images based on a defined kernel. Its widespread use has led to the availability of optimized libraries, making it a practical and efficient choice. For this project, morphological filters from the SimpleITK framework (v2.3.1) were employed.

The main morphological operations utilized in the experiments include:

- **Opening**: A combination of erosion followed by dilation, used to remove small objects or noise while preserving the overall shape of larger structures.
- **Closing**: A combination of dilation followed by erosion, used to fill small holes or gaps in structures while maintaining the shape of larger objects.
- **Hole Filling**: A morphological operation that identifies and fills internal gaps or voids within a binary object, ensuring a continuous structure.

These operations were carefully selected to help address common segmentation artifacts, such as protrusions, insulations, and small holes, with the goal of ultimately improving the accuracy and clarity of the segmentation outputs.

E. Experimental Design

During the experimental phase, a sequential approach was adopted, beginning with experiments aimed at addressing issues with protrusions and insulations, followed by those targeting small holes (i.e. background label insulation within tissue segmentation). Performance was evaluated through both quantitative and qualitative comparisons against the raw (non-post-processed) segmentation. Quantitative evaluation was conducted using the mean Dice coefficient [4] and the mean 95th percentile Hausdorff distance [5] for each segmentation structure. The experiments were carried out on a validation set consisting of 3 subjects, while the final post-processing workflow was assessed on a testing set of 7 subjects to ensure an unbiased evaluation.

For the testing set, in addition to the usual quantitative and qualitative evaluations, a Wilcoxon signed-rank test [6] was conducted to assess whether the metrics of the post-processed segmentations were significantly improved compared to those of the raw (non-post-processed) segmentations. This statistical analysis provides a robust evaluation of the potential improvements achieved through post-processing.

III. Results

A. Experiments on the Validation Set

1) Opening: The initial set of experiments focused on addressing segmentation artifacts such as insulations and protrusions through morphological opening operations. For these experiments, fixed kernel sizes ranging from 1 to 3 were applied globally across all segmentation labels to evaluate their impact on individual structures.

The results demonstrated that certain structures responded differently to kernel size variations. For instance, the Amygdala showed improved quantitative metrics with larger kernel sizes, whereas structures like White Matter performed better with smaller kernel sizes. This variability highlighted the limitations of applying a one-size-fits-all approach to morphological operations.

To overcome this limitation, a data-driven strategy was developed. Instead of hard-coding kernel sizes for specific labels, the topological and geometric characteristics of each structure were leveraged to dynamically assign kernel sizes. Structures with a high surface-to-volume ratio and a high number of fragments, such as Grey Matter, were deemed more "fragile" and assigned a kernel size of 1 to preserve their integrity. Conversely, structures with lower surface-to-volume ratios and fewer fragments, such as the Amygdala, were assigned larger kernel sizes for more aggressive artifact removal.

The categorization process was guided by these key metrics, leading to the development of a data-driven opening (ddO) method that effectively built upon the strengths of the static experiments. This approach demonstrated both quantitative improvements (Figure 2), achieving the highest mean Dice coefficients and lowest mean 95th percentile Hausdorff distances, and qualitative enhancements (Figure 3), with visible removal of insulations and protrusions, ensuring more accurate and visually consistent segmentation results.

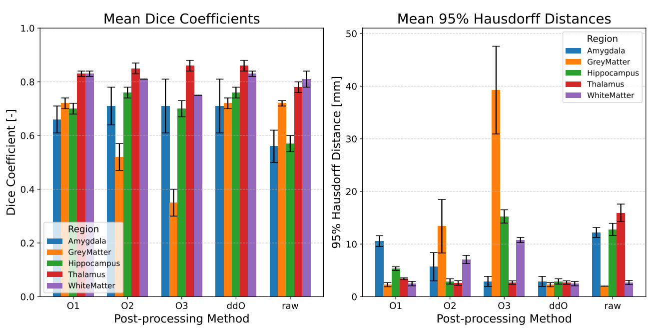

Fig. 2: Quantitative comparison of opening operations: Evaluation of fixed kernel sizes (1 to 3, labeled as O1–O3) compared to the data-driven opening (ddO) and the raw (non-post-processed) segmentation.

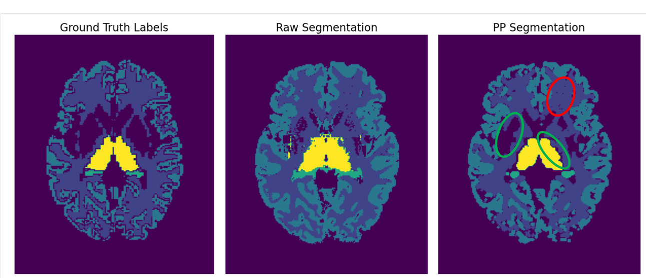

Fig. 3: Qualitative inspection of data-driven opening: A comparison of ground truth labels, raw segmentation, and post-processed (PP) segmentation in the axial view. Green highlights represent successfully removed protrusions and insulations, while red highlights indicate remaining small holes.

2) Closing: Morphological closing was investigated as a method to address small holes in the segmentation. In these experiments, fixed kernel sizes were applied globally across all labels after the application of the ddO method.

The results demonstrated that closing consistently worsened the quantitative metrics compared to ddO alone (Figure 4), with larger kernel sizes leading to increasingly poorer results. Among the tested configurations, the combination of ddO with a closing kernel size of 1 performed the best. This approach successfully filled small holes as intended but also introduced undesired effects, such as the closure of structural gaps like the sulci in the Grey Matter, which negatively impacted the segmentation quality (Figure 5).

Despite the ability to address small holes, the overall decline in performance and the introduction of these unintended artifacts rendered closing unsuitable for the task. Alternative methods were subsequently explored to address the issue of small holes without compromising segmentation quality.

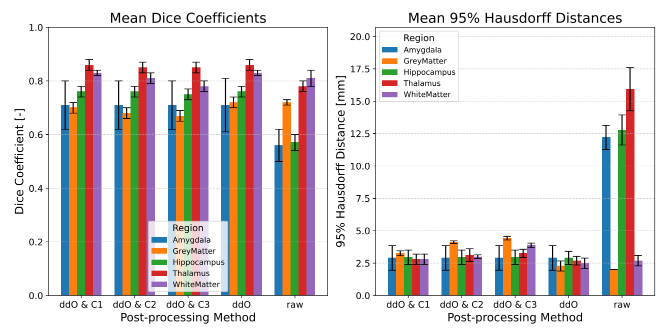

Fig. 4: Quantitative comparison of closing operations: Evaluation of fixed kernel sizes (1 to 3, denoted as C1–C3) following the data-driven opening (ddO), compared to ddO alone and the raw (non-post-processed) segmentation.

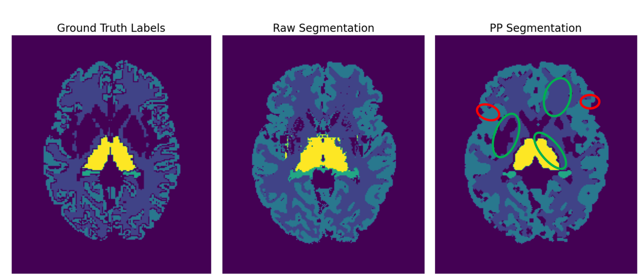

Fig. 5: Qualitative inspection of data-driven opening combined with closing using kernel size 1: A comparison of ground truth labels, raw segmentation, and post-processed (PP) segmentation in the axial view. Green highlights represent successfully removed protrusions, insulations, and small holes, while red highlights indicate the undesired closure of structural gaps.

3) Hole Filling: Geodesic boundary propagation [7] was explored as a method to address the artifacts which presented themselves in the form of holes. However, this method proved unsuitable due to its dependence on the boundary topology.

Specifically, convoluted surfaces such as in the Grey Matter caused the propagation to stop early, hence not filling holes as intended. Additionally, this method lacked the capability to control the size of holes to be filled - an important factor for ensuring traceability of the post-processing by clinicians. Following these insights an approach was adopted which would adhere to the current practices of interpreting brain volumes via 2D slices. The finally implemented method consisted of the following steps: (i) From a per label volume extract one slice and invert back- and foreground. (ii) Perform a connected component analysis to extract all hole candidates. (iii) Threshold the candidates to obtain a list of holes to fill and fill them by labeling them as the current tissue label. (iv) Repeat the process for all slices. The threshold is to be understood as the minimum pixel area of a single hole to not count as artifact, with values of 50, 100 and 150 tested and evaluated both qualitatively and quantitatively. It was found that the quantitative measures of Dice coefficient and Hausdorff distance were unaffected by this processing step while qualitatively the optimal hole filling qualities were found for a threshold of an area of 50 pixels (Figure 6).

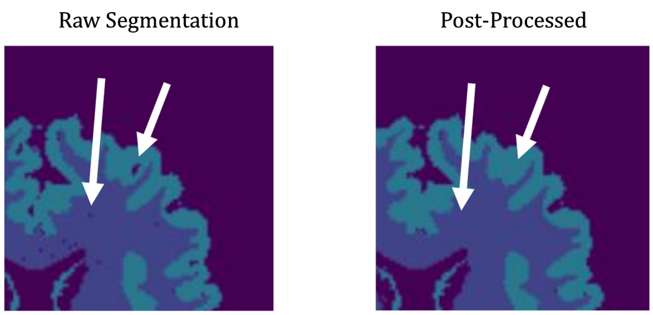

Fig. 6: Qualitative evaluation of the 2D hole filling method with a threshold of 50 pixels. White arrows highlight the removed artifacts.

B. Resulting Post-processing Workflow

The experiments on the validation set resulted in an adaptive post-processing workflow comprising the custom data-driven opening followed by the custom 2D hole filling 50. This combination demonstrated improvements in both quantitative metrics (Figure 7) and qualitative results (Figure 8) on the validation set.

The combined solution effectively addresses key challenges, such as removing insulations and protrusions while successfully filling holes in the segmentation. Although the 2D hole filling step does not significantly affect quantitative metrics, it contributes to a subjectively refined segmentation output.

C. Evaluation on the Testing Set

To ensure an unbiased evaluation, the final post-processing workflow was applied to the testing set. The results demonstrate improvements in both mean Dice and mean 95th Hausdorff metrics (Figure 9), consistent with the outcomes observed during the experiments on the validation set.

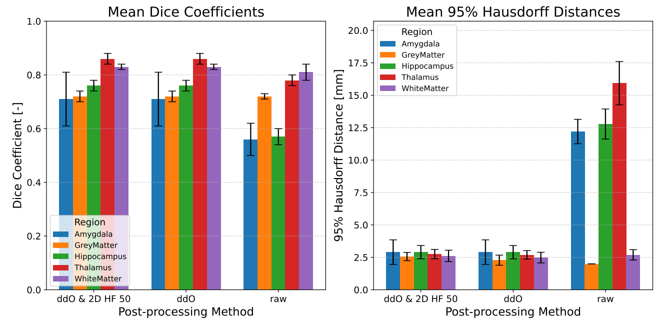

Fig. 7: Quantitative comparison of the resulting post-processing workflow: Assessment of the custom data-driven opening (ddO) followed by 2D hole filling (HF) 50, in comparison to ddO alone and the raw (non-post-processed) segmentation.

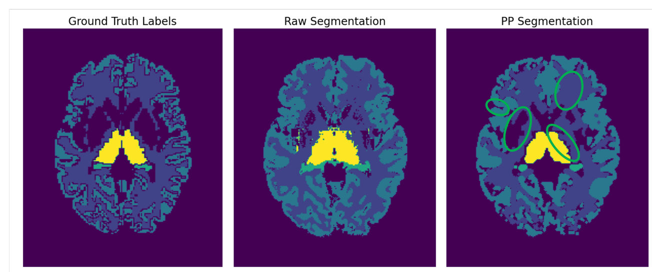

Fig. 8: Qualitative evaluation of the resulting post-processing workflow: A comparison of ground truth labels, raw segmentation, and post-processed (PP) segmentation in the axial view. Green highlights indicate the successful removal of protrusions, insulations, and small holes, while maintaining the sulci in the Grey Matter.

Qualitative inspection (Figure 10) further confirms that the workflow effectively addresses issues such as protrusions, insulations, and small holes, maintaining performance comparable to that observed on the validation set.

A Wilcoxon signed-rank test was performed to evaluate the statistical significance of changes in segmentation accuracy metrics after post-processing. Specifically, the test assessed the increase in the Dice coefficient and the decrease in the Hausdorff distance across all labels, comparing post-processed results to the raw predictions. The test set included seven patients, each with five labeled tissues, resulting in a total of 35 paired samples. The distributions of the raw and post-processed metrics are presented in Figure 11.

The initial significance level was set at 5%; however, after applying the Bonferroni correction, the adjusted significance level was reduced to 2.5%. The metrics, corresponding hypotheses, results, and p-values are summarized in Table 1. The analysis reveals that the adaptive morphological post-processing workflow significantly increases the Dice coefficient and significantly decreases the Hausdorff distance.

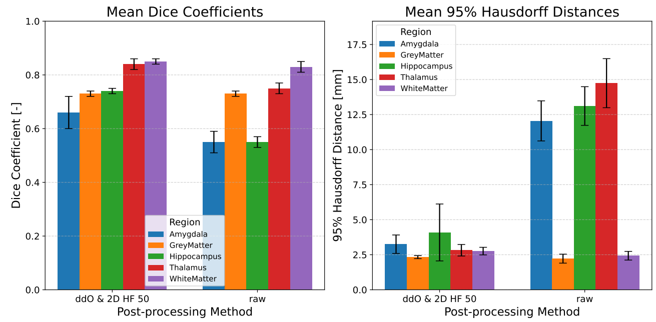

Fig. 9: Quantitative comparison of the resulting post-processing workflow on the testing set: Assessment of the custom data-driven opening (ddO) followed by 2D hole filling (HF) 50, in comparison to the raw (non-post-processed) segmentation.

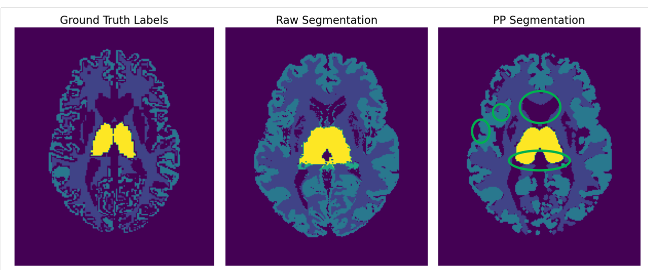

Fig. 10: Qualitative evaluation of the resulting post-processing workflow on the testing set: A comparison of ground truth labels, raw segmentation, and post-processed (PP) segmentation in the axial view. Green highlights indicate the successful removal of protrusions, insulations, and small holes, while maintaining the sulci in the Grey Matter.

IV. Discussion

The use of a per-label data-driven opening and per-slice (2D) hole filling method demonstrated statistically significant improvements in the Dice coefficient and Hausdorff distance. These results support the hypothesis, demonstrating the effectiveness of adaptive morphological post-processing in improving anatomical accuracy. Additionally, qualitative evaluation has shown the methods effectiveness in eliminating noise and spurious artifacts, enhancing visual clarity.

Despite these advancements, several challenges remain. Modifications introduced during post-processing can introduce subjective or unintended biases, potentially distorting the representation of true anatomical structures. Excessive smoothing or filtering also carries the risk of erasing subtle but clinically significant details. Achieving optimal outcomes requires significant engineering and computational effort, and post-processing alone cannot compensate for fundamentally flawed segmentations. Nonetheless, this project has laid the foundation for potential future developments by employing basic data-driven approaches. Refining the algorithms for opening kernel sizes and extending similar improvements to hole-filling thresholds are promising next steps. Furthermore, a valuable

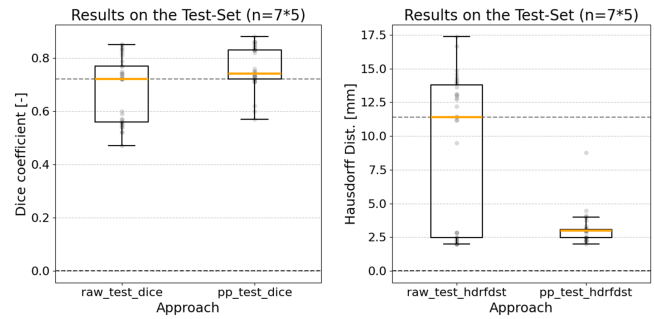

Fig. 11: Qualitative evaluation of the resulting post-processing workflow on the testing set: Distributions of the raw and post-processed (pp) metrics used for the Wilcoxon signed-rank test.

metric	hypothesis	result	p-value
Dice	pp >raw	461	1.3e-6***
Hausdorff	pp <raw	58	3.5e-5***

Tab. 1: Summarized results of the Wilcoxon signed-rank test.

enhancement could involve collaboration with clinicians to enable post-processed results to be overlayed as masks on demand. This would enrich the evaluation toolkit by allowing users to toggle between original and modified representations, offering a more comprehensive view. Insights gained from such experiments could inform the development of more sophisticated post-processing solutions through participatory and interdisciplinary collaboration.

V. Conclusion

The proposed adaptive morphological post-processing workflow, which integrates data-driven opening and 2D hole filling, shows promise in enhancing segmentation accuracy by effectively reducing noise and artifacts. Initial results are encouraging; however, further validation on larger and more diverse datasets, including pathological cases, would yield highly valuable insights. This process should also include testing complementary techniques aligned with clinicians' needs and preferences, as well as optimizing computational efficiency. The overarching goal is to bridge the gap between statistically significant positive results and clinically meaningful improvements in routine healthcare.

While fundamentally flawed segmentations cannot be fully corrected, sustainable digitalization must also be considered. Large models are costly to train and maintain, both financially and in terms of energy consumption. Therefore, if a model delivers anatomically sound predictions with identifiable artifacts, adopting post-processing workflows can be a pragmatic and resource-efficient solution to improve overall outcomes.

In conclusion, a generalizable, multi-step process—such as the one outlined in this work—based on data-driven algorithms could offer significant benefits. This approach holds potential not only for improving outcomes for patients and clinicians but also for addressing broader considerations, such as healthcare costs and environmental sustainability, ultimately benefiting society as a whole.

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